



Name : Mrs. VRISHALI SINGH
 Age / Gender : 46 Year(s)/ Female
 Contact No. : +919892932443
 Address : SOLITAIRE 2, FLAT NO 201, OPPOSITE KAL...
 Pin code : 411045

VID No. : 260176516582543
 PID No. : P18126566925757
 Referred by : SELF
 Registered On : 06/04/2026 2:44 PM
 Collected On : 06/04/2026 2:42 PM
 Reported On : 06/04/2026 9:27 PM

Investigation

Observed Value

Unit

Biological Reference Interval

Iron Studies

(Serum)

Iron, Serum

(FerroZine)

51

µg/dL

33-193

UIBC

(FerroZine)

366

µg/dL

120-470

TIBC

(Calculated)

417

µg/dL

250-450

Transferrin Saturation

(Calculated)

12

%

14-50

Interpretation :

1. Measurements of serum iron, TIBC and the percentage of iron saturation of transferrin are useful screening tests for iron deficiency anaemia.
2. However, serum iron exhibits significant diurnal variation and may transiently rise or reach reference values after dietary or iron supplements & post blood transfusion.
3. The diagnostic specificity of a low serum iron for iron deficiency is lost in the presence of acute & chronic inflammatory processes as the concentrations of iron and transferrin in the serum are significantly affected, and fall rapidly as part of the acute phase response irrespective of the iron stores status in the body.
4. Hence, Concurrent measurement of the markers mentioned in the below interpretative table alongwith serum iron studies improves the diagnostic specificity for iron deficiency anaemia & also provides a reliable work up for microcytic hypochromic anaemia.

Tests	Iron Deficiency anaemia	Anaemia of Chronic disease	Iron overload	Hemoglobinopathy (Especially Trait)
Serum Iron	Decreased	Decreased	Increased	Normal
Serum Total Iron Binding Capacity	Increased	Decreased or Normal	Increased or Normal	Normal
% Transferrin Saturation	Decreased	Decreased or Normal	Increased or Normal	Normal
Serum UIBC	Increased	Decreased or Normal	Decreased	Normal
Serum Ferritin	Decreased	Increased	Increased or Normal	Normal
Serum Soluble Transferrin receptor	Increased	Normal	Decreased	Normal
Serum Hepcidin	Normal	Increased	Normal	Normal

Associated Tests :

1. Serum Soluble Transferrin receptor
2. Serum Hepcidin

Interpretation :

Dr. MANGESH BOLEGAVE
 M.D. Pathology
 Consultant Pathologist
 Reg no. 2001/02/0346

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Investigation**Observed Value****Unit****Biological Reference Interval**

- Measurements of serum iron, UIBC-Unsaturated Iron binding Capacity and the percentage of iron saturation of transferrin are useful screening tests for iron deficiency anemia.

Tests	Iron Deficiency anemia	Anemia of Chronic disease	Iron overload	Hemoglobinopathy (Especially Trait)
Serum Iron	Decreased	Decreased	Increased	Normal
Serum Total Iron Binding Capacity	Increased	Decreased or Normal	Increased or Normal	Normal
% Transferrin Saturation	Decreased	Decreased or Normal	Increased or Normal	Normal
Serum UIBC	Increased	Decreased or Normal	Decreased	Normal
Serum Ferritin	Decreased	Increased	Increased or Normal	Normal
Serum Soluble Transferrin receptor	Increased	Normal	Decreased	Normal
Serum Hepcidin	Normal	Increased	Normal	Normal

- The sum of the serum iron and UIBC represents total iron-binding capacity (TIBC) and is a measurement for the maximum iron concentration that the iron complex can bind.

Clinical Utility:

- Iron measurements are used in the diagnosis of iron deficiency anemia, hemochromatosis, microcytic anemia, macrocytic anemia, chronic renal disease, erythropoietin deficiency, hemolytic anemia and hemoglobinopathy. The test is also undertaken in patients of chronic renal disease, hemolytic anemia and hemoglobinopathy.

Caution:

- Serum iron exhibits significant diurnal variation and may transiently rise or reach reference values after dietary or iron supplements & post blood transfusion.
- Patients treated with Deferoxamine containing drugs may give low values with the UIBC assay.

Associated Tests:

Soluble Transferrin Receptor Serum (S0036), Serum Hepcidin (H0077).Serum Ferritin (F0018)

Reference:

- Package insert
- Wallach's interpretation of diagnostic tests, Ed11, 2020
- Henry's Clinical Diagnosis and Management by Laboratory Methods. 23rd ed; 2017.
- Tietz fundamentals of clinical chemistry 6th edition. Burtis CA, Ashwood ER, Bruns DE, 2008.

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Investigation**Observed Value****Unit****Biological Reference Interval****Glucose Fasting**

(Fluoride Plasma - F, Hexokinase)

99

mg/dL

Normal: 70-99

Impaired Fasting Glucose(IFG): 100-125

Diabetes mellitus: ≥ 126 (on more than one occasion)
(American diabetes association guidelines 2022)**Amylase, Serum**

(Serum, Enzymatic colorimetric Assay - IFCC)

42

U/L

25-125

Interpretation:

- High levels are seen in various pancreatic as well as salivary gland disorders, intestinal blockage, peptic ulcer, appendicitis, viral hepatitis, burns and acute alcohol poisoning.
- Low levels are seen in bone fracture, chronic heart failure, chronic pancreatitis, liver and kidney diseases.

Clinical Utility:

- Helps in diagnosing acute pancreatitis and other pancreatic diseases.
- In acute pancreatitis, high amylase levels are usually associated with high lipase concentrations, although lipase levels may take a while to rise than blood amylase levels and will remain elevated for a longer time period.

Note:

- Pregnancy and recent kidney transplant affects the test results.
- Usage of drugs like aspirin, diuretics, oral contraceptives, corticosteroids, indomethacin, ethyl alcohol and opiates also interfere in test results.
- Amylase levels may be increased in patients with Macroamylase. It can be confirmed by testing serum lipase and urinary amylase levels.

Associated Tests:

Lipase Serum (L0068), Urinary amylase (A0433_24/ A0433_24H)

Reference:

- Kit Insert.
- Henry's Clinical Diagnosis and Management by Laboratory Methods. 24th ed. Philadelphia, PA: Elsevier; 2022:chap 23

SGOT (AST)

(Serum, Enzymatic)

18.8

U/L

0-32

SGPT (ALT)

(Serum, Enzymatic)

13.6

U/L

0-33

Bilirubin Total, Direct, Indirect

(Serum)

Bilirubin Total

(Diazo method)

0.36

mg/dL

0-1.2

Bilirubin Direct

(Diazo method)

0.15

mg/dL

0.0-0.3

Bilirubin- Indirect

(Calculated)

0.21

mg/dL

0.1-1.0

Proteins, Serum

(Serum)

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Investigation	Observed Value	Unit	Biological Reference Interval
Total Protein (Biuret test)	8.4	g/dL	6.4-8.3
Albumin, Serum (Bromocresol Green (BCG))	4.7	gm/dL	3.5-5.2
Globulin (Calculated)	3.70	gm/dL	1.8-3.6
Albumin/Globulin Ratio	1.27		1.1-2.2

Interpretation:

- Total Proteins are useful in the diagnosis and treatment of disease involving liver, kidney, bone marrow ,metabolic and nutritional disorders
- The protein concentration of serum is an indicator of the hydration state of the body. Prolonged bed rest results in decreased total protein concentration.
- The A/G ratio measures the relative ratio of albumin to globulin
- Low A/G ratio may indicate viral infections, liver and kidney disease, or autoimmune disorders. These diseases increase globulin and decrease albumin thus lowering the A/G ratio.
- A high A/G ratio may indicate diseases that make the body produce less globulin, such as genetic disorders or may result from the use of immunosuppressive drugs.

Reference:

- Juraschek SP, Moliterno AR, Checkley W, Miller ER 3rd. The Gamma Gap and All-Cause Mortality. PLoS One. 2015 Dec 2;10(12):e0143494
- Busher JT. Serum Albumin and Globulin. In: Walker HK, Hall WD, Hurst JW, editors. Clinical Methods: The History. Physical, and Laboratory Examinations. 3rd edition. Boston: Butterworths; 1990. Chapter 101.
- Pack Insert.



Alkaline Phosphatase, Serum (Serum, p-nitrophenyl phosphate (pNPP))	129	U/L	35-104
Gamma GT (GGTP) (Serum, Enzymatic IFCC)	31	U/L	< 40
LDH (Lactate Dehydrogenase), Serum (Serum)	212	U/L	0-250

Interpretation:

- Lactate dehydrogenase (LDH) is an enzyme that is present in heart, liver, muscle and kidney tissues.
- Very high levels of LDH are seen in patients with megaloblastic anaemia, carcinoma and shock.
- Moderate increase seen in muscular disorders, nephrotic syndrome, and cirrhosis.
- Mild increases in LDH activity are seen in cases of myocardial or pulmonary infarction, leukemia, hemolytic anemia and non-viral hepatitis
- Many cancers cause a general increase in LDH levels or an increase in one of its isozymes.

Clinical Utility:

- It is used to screen for tissue damage along with other related tests like LDH isoenzymes.
- It is also used to monitor severe infections or conditions like hemolytic or megaloblastic anemias, kidney disease, and liver disease

Associated Test :

CPK Total Serum (C0165), SGPT ALT Serum (S0019), Troponin-I, Serum(T0128), Troponin-T, Serum (T0129) ,

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Biological Reference Interval

LDH Isoenzymes (L0016)

Reference :

- Package Insert
- Klein R, Nagy O, Tóthová C, Chovanová F. Clinical and Diagnostic Significance of Lactate Dehydrogenase and Its Isoenzymes in Animals. Vet Med Int. 2020 Jun 15;2020:5346483.



BUN, Serum

(Serum, Urease)

11.0

mg/dL

6-20

Remark: In blood, Urea is usually reported as BUN and expressed in mg/dl. BUN mass units can be converted to urea mass units by multiplying by 2.14.



Uric Acid, Serum

(Serum, Uricase)

5.2

mg/dL

2.4-5.7



Electrolytes, Serum

(Serum, Ion Selective Electrode (ISE))

Sodium, Serum

139

mmol/L

136-145

Interpretation:

- Low levels are noted in prolonged vomiting or diarrhea, diminished reabsorption in the kidney and excessive fluid retention. High levels are seen in case of excessive fluid loss, high salt intake and increased kidney reabsorption

Potassium, Serum

3.9

mmol/L

3.5-5.1

Interpretation:

- Low levels are noted in reduced intake of dietary potassium or excessive loss of potassium from the body due to diarrhea, prolonged vomiting or increased renal excretion. High levels may be caused by dehydration or shock, severe burns, hemolysis, diabetic ketoacidosis, and retention of potassium by the kidney

Chloride, Serum

100

mmol/L

98-107

Interpretation:

- Low levels** are noted in reduced dietary intake, prolonged vomiting and reduced renal reabsorption as well as some forms of acidosis and alkalosis. High levels are found in dehydration, kidney failure, some forms of acidosis, high dietary or parenteral chloride intake, and salicylate poisoning.



GFR with Creatinine (Above 18 Years)

(Serum, Jaffes method)

Creatinine, Serum

0.68

mg/dL

0.60-1.10

eGFR (CKD-EPI)

109

mL/min/1.73m²

Normal OR high: >=90
 ,Normal OR high: >=90
 ,Mild decrease :60-89
 ,Mild moderate decrease:45-59
 ,Moderate to severe decrease:30-44
 ,Severe decrease:15-29
 ,Kidney failure:<15

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Equation is not valid for patients below 18 years of age. Calculated by IDMS-Traceable CKD-EPI creatinine equation.

Interpretation:

Kidneys filter the blood by removing waste and extra water to make urine. The glomerular filtration rate (GFR) shows how well the kidneys are filtering. When found early, important steps can be taken to protect their kidneys.

In adults, the normal eGFR number is usually more than 90. eGFR declines with age, even in people without kidney disease. See chart below for average estimated eGFR based on age.

Age (years)	Average eGFR
20-29	116
30-39	107
40-49	99
50-59	93
60-69	85
70+	75

Note:

- Race-'OTHERS' means other than black (Caucasian / Asians)
- Creatinine test done by IDMS traceable kit
- Patient flyer https://www.kidney.org/sites/default/files/01-10-8374_2210_patflyer_egfr.pdf

**Lipid Profile - 2 (Mini - Fasting)**

(Serum)

Cholesterol Total, Serum

(Enzymatic)

187

mg/dL

Desirable: < 200

Borderline High: 200-240

High: >= 240

Triglycerides, Serum

(Enzymatic)

149

mg/dL

Normal: < 150

Borderline High: 150-199

High: 200-499

Very High: >= 500

HDL Cholesterol Direct

(Homogeneous Enzymatic colorimetric Assay)

58

mg/dL

> 40

Non HDL Cholesterol

129.00

mg/dL

Optimal:<130

,Desirable:130-159

,Borderline high:159-189

,High:189-220

,Very High:>=220

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Investigation	Observed Value	Unit	Biological Reference Interval
LDL Cholesterol	99.20	mg/dL	Optimal:<100 ,Near Optimal:100-129 ,Borderline high:130-159 ,High:160-189 ,Very High:>=190
VLDL Cholesterol	29.80	mg/dL	6-38
LDL/HDL Ratio	1.71		2.5-3.5
Cholesterol / HDL Ratio	3.22		3.5-5.0

Note:

Reference Interval as per National Cholesterol Education Program (NCEP) Adult Treatment Panel III Report.

VLDL, CHOL/HDL RATIO, LDL/HDL RATIO, LDL Cholesterol, serum, Non HDL Cholesterol are calculated parameters



Apolipoproteins B/A1

(Serum)

Apolipoproteins A1 (Immunoturbidimetry)	151	mg/dL	108-225
Apolipoproteins B (Immunoturbidimetry)	95.9	mg/dL	60-117
Apolipoprotein B/A1 Ratio	0.64		0.30-0.90

Assay of Apolipoproteins A-1 and B is helpful in assessing the risk of atherosclerosis and has greater prognostic power than the sole determination of HDL and LDL cholesterol. A powerful parameter for estimating the risk of atherosclerosis is the quotient Apo B/Apo A-1. The higher the quotient the greater the risk of atherosclerosis.



HsCRP (High Sensitivity CRP)

(Serum, Immunoturbidimetry)

20.21

mg/L

Low: < 1.0

Average: 1.0-3.0

High: > 3.0

Interpretation:

1. High sensitivity C (Hs-CRP) reactive protein (hs CRP) helps in identification of risk for cardiovascular diseases
2. It also helps in monitoring the prognosis in patients with cardiovascular diseases such as stable coronary disease or acute coronary syndromes.
3. It is also raised in active infection, systemic inflammatory processes or tissue injury, Malignancy, burns and ongoing hormonal therapy like estrogen and progesterone.
4. It also increases after 24-48 hours of acute myocardial infarction, peaks at 72 hours and comes down after several weeks.
5. Decreased levels are seen in Exercise, weight loss, moderate alcohol consumption and drugs (eg. statins, fibrates, niacin).

Caution:

Patients with evidence of active infection, systemic inflammatory processes or trauma should not be tested for Hs-CRP for cardiovascular indications, since it may show false high results

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Investigation**Observed Value****Unit****Biological Reference Interval****Cardiovascular Risk Classification by HsCRP***

Risk Level	HsCRP (Mg/L)
Low	<1.0
Average	1.0-3.0
High	>3.0

*Cardiovascular disease risk assessment guidelines recommended by the CDC and the American Heart Association (CDC/AHA)

Associated Tests:

Cardiac Injury Profile-Mini (C0034).

Reference:

- Package insert
- Wallach's interpretation of diagnostic tests, Ed11, 2020
- Sehring SM, Goyal A, Patel BC. C Reactive Protein. [Updated 2021 Dec 28]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK441843/>
- Henry's Clinical Diagnosis and Management by Laboratory Methods. 23rd ed; 2017.
- Tietz fundamentals of clinical chemistry 6th edition. Burtis CA, Ashwood ER, Bruns DE, 2008.

**CPK Total**

90

U/L

< 170

(Serum, Enzymatic)

Interpretation:

- Creatinine Kinase (CK) is also called as creatinine phosphokinase (CPK) or phosphocreatine kinase.
- CPK levels are useful for diagnosing and monitoring of Heart & Muscle related issues such as myocardial infarction (MI) and myopathies.
- Creatine kinase activity begins to rise within 12 hours of Acute Myocardial infarction symptoms, peaks at 24 to 36 hours, and normalizes after 48 to 72 hours

Clinical Utility:

- The CPK test used primarily To help diagnose a heart attack/chest pain To assess the extent of damage to heart or muscle tissue To determine muscular dystrophy.It also helps in diagnosis of Myositis Malignant hyperthermia Other conditions related to muscle breakdown.

Caution:

- Exercise, muscle trauma, recent surgery, cardiac catheterization, some vaccines can elevate CPK values.
- Presence of Macro CK may elevate CPK levels.
- Medications that interfere with CPK test are steroids, anesthetics, Amphotericin B, alcohol, Cocaine.

Associated Tests:

- CPK Isoenzymes Serum (C0164).

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- Tietz fundamentals of clinical chemistry 6th edition. Burtis CA, Ashwood ER, Bruns DE, 2008.
- Aujla RS, Patel R. Creatine Phosphokinase. [Updated 2022 Apr 14]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK546624/>



Calcium, Serum

(Serum, NM-BAPTA)

9.83

mg/dL

8.6-10.0



Phosphorus, Serum

(Serum, Molybdate UV)

3.8

mg/dL

2.5-4.5



RA (Rheumatoid Arthritis) Factor, Serum

(Serum, Immunoturbidimetry)

13.4

IU/mL

Non-reactive: < 14

Please note change in Reference range and method

Interpretation

- Increased in RA, Chronic hepatitis, chronic viral infections, cirrhosis, Infectious mononucleosis, leishmaniasis, leprosy, malaria sarcoidosis Syphilis, TB etc.
- High levels are also seen in inflammatory rheumatic diseases and in various non rheumatic diseases.

Clinical Utility:

- The detection of rheumatoid factor (RF) is one of the criteria of the American Rheumatism Association (ARA) for the diagnosis of Rheumatoid Arthritis (RA) especially when clinical diagnosis is difficult.
- RFs play an important role in the differential diagnosis between RA and other rheumatic diseases. They also permit prognostic statements with regards to RA.

Note:

- Values of RA may increase with age. It should be interpreted along with overall clinical picture.
- Recent blood transfusion, multiple vaccinations or transfusions may affect results.
- Serum with cryoglobulin or high lipid levels may cause false-positive test results.

Associated Tests:

Rheumatoid Arthritis Panel (R0022), Cyclic Citrullinated Peptide Antibody (CCP Antibody) Serum (C0047)

Reference:

- Package insert
- Wallach's interpretation of diagnostic tests, Ed11, 2020
- Henry's Clinical Diagnosis and Management by Laboratory Methods. 23rd ed; 2017.
- Tietz fundamentals of clinical chemistry 6th edition. Burtis CA, Ashwood ER, Bruns DE, 2008.



CRP (C Reactive Protein), Serum

(Serum, Immunoturbidimetry)

21.57

mg/L

< 5.0

Interpretation :

C-Reactive Protein (CRP) is an acute phase reactant which is a protein released during inflammation by cytokines.

Increased CRP:

Seen in various inflammatory conditions such as infection (bacterial/ viral/ fungal), autoimmune diseases, Cardiovascular disease, Inflammatory bowel disease, injuries etc

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Investigation**Observed Value****Unit****Biological Reference Interval****Decreased CRP:**

Seen in Patients on carboxypenicillins, Liver failure etc.

Limitations:

- Elevated CRP values are non-specific and should not be interpreted without a complete history.
- Heterophile antibodies may give falsely high levels.
- Elevated CRP can also be genetic, age related and sedentary lifestyle, exposure to environmental toxins, diet that includes refined, processed manufactured foods.

Note :

If the test has been ordered for COVID-19 purpose, you may take one of the following profiles for further investigation under your clinician's advice.

- Covid Monitor Initial profile (C0374) from Day 1 to Day 5
- Covid Monitor maintenance profile (C0375) from Day 5 to Day 10
- Covid Monitor Recovery profile (C0376) after discharge.

Associated Tests:

Interleukin -6 CLIA Plasma (I0279)

Reference:

- Package insert
- Wallach's interpretation of diagnostic tests, Ed11, 2020
- Henry's Clinical Diagnosis and Management by Laboratory Methods. 23rd ed; 2017.
- Khera a, Mcguire DK, Murphy, et al. Race and gender differences in C- reactive protein levels. J Am Coll Cardiol. 2005;46:464-469.
- Tietz fundamentals of clinical chemistry 6th edition. Burtis CA, Ashwood ER, Bruns DE, 2008.

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Routine Examination Profile - Urine

Investigation

Observed Value

Unit

Biological Reference Interval

GENERAL EXAMINATION

(Urine)

Colour	Pale Yellow		Pale Yellow
Transparency (Appearance)	Clear		Clear
Reaction (pH) (Ph Indicator)	6.5		4.5-8
Specific Gravity (Refractometer method)	1.010		1.005-1.025

CHEMICAL EXAMINATION (AUTOMATED URINE CHEMISTRY)

(Urine)

Urine Protein (Albumin)	Absent		Absent
Urine Ketones (Acetone)	Absent		Absent
Urine Glucose (Sugar)	Absent		Absent
Urobilinogen	Normal		Normal
Bilirubin	Negative		Negative
Nitrite	Negative		Negative
Leucocyte Esterase	+		Negative

MICROSCOPIC EXAMINATION (CUVETTE BASED IMAGING TECHNOLOGY)

(Urine)

Red blood cells	0.3	/hpf	0-2
Dysmorphic Red Blood Cells	Absent		Absent
Pus cells (WBCs)	0.9	/hpf	0-5
Epithelial cells	2.5	/hpf	0-5
Crystals	0.0		0-1.36
CRY - Calcium-oxalate monohydrate	0.0		0-1.36
CRY - Calcium-oxalate dihydrate	0.0		0-1.36
CRY - Triple-phosphate	0.0		0-1.36
CRY - Uric acid	0.0		0-1.36
Casts - Hyalin	0.0		0-2
Casts - Pathological	0.0		0-1.36
Bacteria	Absent		Absent
Trichomonas Vaginalis	Absent		Absent
Yeast cells	Absent		Absent

Dr. MANGESH BOLEGAVE
 M.D. Pathology
 Consultant Pathologist
 Reg no. 2001/02/0346

MEDICAL LABORATORY REPORT



260176516582543

Name : Mrs. VRISHALI SINGH
Age / Gender : 46 Year(s)/ Female
Contact No. : +919892932443
Address : SOLITAIRE 2, FLAT NO 201, OPPOSITE KAL...
Pin code : 411045

VID No. : 260176516582543
PID No. : P18126566925757
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Investigation**Observed Value****Unit****Biological Reference Interval**

1. Urine routine and microscopy is a screening test.
2. Abnormal results of chemical examination are confirmed by manual methods.
3. Chemical examination through Dipstick includes test methods as Protein (Protein Error Principle), Glucose (Glucose Oxidase-Peroxidase), Ketone (Legals Test), Bilirubin (Azo- Diazo reaction), Urobilinogen (Diazonium ion Reaction) Nitrite (Griess Method).
4. All abnormal results of chemical examination are confirmed by manual methods. Protein by sulfosalicylic acid method, Glucose by Benedict's method, Ketone by Rothera's method, Bile salt by Sulfur granule method, Bile pigment by Fouchet method, Urobilinogen by Ehrlich Method.
5. Pre-test conditions to be observed while submitting the sample- First void, mid-stream urine, collected in a clean, dry, sterile container is recommended for routine urine analysis, avoid contamination with any discharge from vaginal, urethra, perineum, as applicable, avoid prolonged transit time & undue exposure to sunlight.
6. During interpretation, points to be considered are Negative nitrite test does not exclude the presence of the bacteria or urinary tract infections.
7. Trace proteinuria can be seen with many physiological conditions like prolonged recumbency, exercise, high protein diet etc.
8. False reactions for bile pigments, proteins, glucose and nitrites can be caused by peroxidase like activity by disinfectants, therapeutic dyes, ascorbic acid and certain drugs etc.
9. Physiological variations may affect the test results.
10. The Microscopic examination findings reported are in decimal numbers as they represent arithmetic mean of multiple fields scanned using Microscopy.

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CBC Haemogram

Investigation	Observed Value	Unit	Biological Reference Interval
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Erythrocytes

Haemoglobin (Hb)	10.8	gm/dL	12.0-16
Erythrocyte (RBC) Count	5.19	mill/cu.mm	4.2-5.4
PCV (Packed Cell Volume)	34.9	%	37-47
MCV (Mean Corpuscular Volume)	67.2	fL	82-101
MCH (Mean Corpuscular Hb)	20.8	pg	27-34
MCHC (Mean Corpuscular Hb Conc.)	31.0	gm/dL	31.5-36
RDW (Red Cell Distribution Width)	17.0	%	11.5-14.0

RBC Morphology

Hypochromia	++
Microcytosis	++
Anisocytosis	+

Leucocytes

Total Leucocytes (WBC) Count	7100	cells/cu.mm	4300-10300
Absolute Neutrophils Count	3720	cells/cu.mm	2000-7000
Absolute Lymphocyte Count	2599	cells/cu.mm	1000-3000
Absolute Monocyte Count	476	cells/cu.mm	200-1000
Absolute Eosinophil Count	249	cells/cu.mm	20-500
Absolute Basophil Count	57	cells/cu.mm	20-100
Neutrophils	52.4	%	40-80
Lymphocytes	36.6	%	20-40
Monocytes	6.7	%	2.0-10
Eosinophils	3.5	%	1-6
Basophils	0.8	%	0-2

Platelets

Platelet count	306	1000/c.mm	140-440
MPV (Mean Platelet Volume)	8.2	fL	7.8-11
PCT (Platelet Haematocrit)	0.250	%	0.2-0.5
PDW (Platelet Distribution Width)	16.3	%	9-17

EDTA Whole Blood - Tests done on Automated Five Part Cell Counter. (Haemoglobin by photometric measurement, WBC, RBC Platelet count by impedance method, WBC differential by VCS technology other parameters calculated) All Abnormal Haemograms are reviewed confirmed microscopically. Differential count is based on approximately 10,000 cells.

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Investigation**Observed Value****Unit****Biological Reference Interval****ESR (Erythrocyte Sedimentation Rate)**

17

mm/hr

0-20

(EDTA Whole Blood)

Method: Automated Westergren**Interpretation:**

1. It indicates presence and intensity of an inflammatory process, never diagnostic of a specific disease. Changes are more significant than a single abnormal test.
2. It is a prognostic test and used to monitor the course or response to treatment of diseases like tuberculosis, bacterial endocarditis, acute rheumatic fever, rheumatoid arthritis, SLE, Hodgkins disease, temporal arteritis, polymyalgia rheumatica.
3. It is also increased in pregnancy, multiple myeloma, menstruation, and hypothyroidism.

**HbA1c- Glycated Haemoglobin**

(EDTA Whole Blood)

HbA1C- Glycated Haemoglobin

6.5

%

Non-diabetic: <= 5.6

Pre-diabetic: 5.7-6.4

Diabetic: >= 6.5

Refer interpretation for monitoring ranges.

(High-Performance Liquid Chromatography (HPLC))

Estimated Average Glucose (eAG)

140

mg/dL

Interpretation & Remark:

1. HbA1c is used for monitoring diabetic control. It reflects the estimated average glucose (eAG).
2. HbA1c has been endorsed by clinical groups & ADA (American Diabetes Association) guidelines 2017, for diagnosis of diabetes using a cut-off point of 6.5%.
3. Trends in HbA1c are a better indicator of diabetic control than a solitary test.
4. Low glycated haemoglobin (below 4%) in a non-diabetic individual are often associated with systemic inflammatory diseases, chronic anaemia (especially severe iron deficiency & haemolytic), chronic renal failure and liver diseases. Clinical correlation suggested.
5. To estimate the eAG from the HbA1C value, the following equation is used: $eAG(mg/dl) = 28.7 * A1c - 46.7$
6. Interference of Haemoglobinopathies in HbA1c estimation.
 - A. For HbF > 25%, an alternate platform (Fructosamine) is recommended for testing of HbA1c.
 - B. Homozygous hemoglobinopathy is detected, fructosamine is recommended for monitoring diabetic status
 - C. Heterozygous state detected (D10/ Tosho G8 is corrected for HbS and HbC trait).

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MEDICAL LABORATORY REPORT**Sample Collected At:**

Call2 Connect Pvt Ltd, 4th Dimension Raheja Mindspace Malad
 West Mumbai 400086

Processing Location :

Processing Location Metropolis Healthcare Ltd. Bhandarkar
 Road, Pune - 411004



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Investigation**Observed Value****Unit****Biological Reference Interval**

7. In known diabetic patients, following values can be considered as a tool for monitoring the glycemic control.
- Excellent Control - 6 to 7 %,
 - Fair to Good Control - 7 to 8 %,
 - Unsatisfactory Control - 8 to 10 %
 - and Poor Control - More than 10 % .

Note : Hemoglobin electrophoresis (HPLC method) is recommended for detecting hemoglobinopathy.

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Investigation

Observed Value

Unit

Biological Reference Interval

Vitamin B12 (Cyanocobalamin)

515

pg/mL

197-771

(Serum, Electrochemiluminescence immunoassay (ECLIA))

Interpretation :

1. Vit B12 levels are decreased in megaloblastic anemia, partial/total gastrectomy, pernicious anemia, peripheral neuropathies, chronic alcoholism, senile dementia, and treated epilepsy.
2. An associated increase in homocysteine levels is an independent risk marker for cardiovascular disease and deep vein thrombosis.
3. Holo Transcobalamin II levels are a more accurate marker of active VitB12 component.

Caution:

- Patients on Biotin supplement may have interference in some immunoassays. With individuals taking high dose Biotin (more than 5 mg per day) supplements, at least 8-hour wait time before blood draw is recommended.

Disclaimer:

- High levels of Vitamin B12 may be due to exogenous supplementation. Kindly correlate clinically.

Associated Tests

- Active Vitamin B12 (V0012), Homocysteine reflex Vitamin B12-folate serum (H0310), Homocysteine Serum (H0254), RBC Folate R0007.

Reference:

1. Package insert
2. Arch Pathol Lab Med—Vol 141, November 2017

Homocysteine

17.4

μmol/L

4.44-13.56

(Serum, Enzymatic)

Interpretation:

- 10.70 mol/L 5-15 Increased levels of homocysteine are seen in hyperhomocysteinemia or homocystinuria (homocysteine excreted in urine) due to nutritional and genetic deficiencies.
- Levels are also increased in various diseases like cancers of ovary or breast, Leukaemia, chronic liver or renal failure, post-menopausal state, drug usage, alcohol consumption, and cigarette smoking
- High levels of homocysteine can also lead to blood clots or blood vessel blockages
- Maternal hyperhomocysteinemia is associated with increased risk of pregnancy complications.

Clinical Utility:

- Homocysteine values can help in the diagnosis and treatment of patients suspected of having hyperhomocystinemia and homocystinuria.
- Homocysteine is a marker for risk assessment of coronary artery disease (CAD), stroke and deep vein thrombosis.

Caution:

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Investigation**Observed Value****Unit****Biological Reference Interval**

Drugs like methotrexate, carbamazepine, phenytoin, nitrous oxide, anticonvulsants and 6-azauridine triacetate increase levels of homocysteine.

Reference:

- Kit Insert
- Maron BA, Loscalzo J. The treatment of hyperhomocysteinemia. Annu Rev Med. 2009;60:39-54

**Thyroid Profile - 2**

(Serum , Electrochemiluminescence immunoassay (ECLIA))

FT3 (Free Triiodothyronine)**5.20**

pg/mL

2.0-4.4

First Trimester : 2.46 - 3.49

Second Trimester : 2.09 - 3.55

Third trimester : 2.01 - 3.27

Interpretation :

Total T3 & T4 values may also be altered in other conditions due to changes in serum proteins or binding sites Pregnancy, Drugs (Androgens, Estrogens, O C Pills , Phenytoin), Nephrosis etc. In such cases Free T3 and Free T4 give corrected values.

FT4 (Free Thyroxine)

1.11

ng/dL

0.93-1.7

First Trimester : 0.7-2.0

Second Trimester : 0.5-1.6

Third Trimester : 0.5-1.6

Interpretation :

Total T3 & T4 values may also be altered in other conditions due to changes in serum proteins or binding sites Pregnancy, Drugs (Androgens, Estrogens, O C Pills , Phenytoin), Nephrosis etc. In such cases Free T3 and Free T4 give corrected values.

TSH (Thyroid Stimulating Hormone) -**0.416**

μIU/mL

0.54-5.3

Ultrasensitive, Serum

First Trimester : 0.33-4.59

Second Trimester : 0.35-4.10

Third trimester : 0.21-3.15

Medical Remarks: Advised : Thyroid antibodies (T0061).

Interpretation :

- Increased TSH is seen with intake of Iodine, Lithium, Amiodarone drugs and also indicates considerable physiologic & seasonal variation.
- Decreased TSH values require correlation with patient age & clinical symptoms and seen with intake of few drugs e.g. L-dopa, glucocorticoids.
- Transient alteration in TSH is seen in non-thyroidal illness like severe infections, liver disease, renal and heart failure, severe burns, trauma and surgery etc.

Clinical Utility: Levels of TSH are used for monitoring of thyroid related disorders.

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Investigation

Observed Value

Unit

Biological Reference Interval

Caution: Patients on Biotin supplement may have interference in some immunoassays. For sample collection, at least 8-hours wait time is recommended for individuals taking high dose of Biotin (more than 5 mg per day) supplements.

Note: TSH levels may fluctuate based on few factors such as pregnancy, illness and age. Also, time of sample collection, technologies used to analyze the test, usage of certain drugs, diet may have impact on TSH levels. TSH may show around 50% variation even when done at different times of day due to its association with circadian rhythm.

Associated Tests: T3 (T0029), T4 (T0031) free T3 (T0028), free T4 (T0030), reverse T3 (R1004), Thyroid Antibodies (T0061), Thyroid Comprehensive Profile-1 (T0062)

Reference:

- Clinical Chemistry 50:12, 2338-2344 (2004) and Ind J Clin Biochem (Apr-June 2014) 29(2):189-195.
- Ref: Arch Pathol Lab Med—Vol 141, November 2017.
- Fisher DA. Physiological variations in thyroid hormones: physiological and pathophysiological considerations. Clin Chem. 1996 Jan;42(1):135-9. PMID: 8565215

INTERPRETATION

TSH	T3 / FT3	T4 / FT4	Suggested Interpretation for the Thyroid Function Tests Pattern
Within Range	Decreased	within Range	•Isolated Low T3-often seen in elderly & associated Non-Thyroidal illness. In elderly the drop in T3 level can be upto 25%.
Raised	Within Range	Within Range	•Isolated High TSH especially in the range of 4.7 to 15 mIU/ml is commonly associated with Physiological & Biological TSH Variability. •Subclinical Autoimmune Hypothyroidism •Intermittent T4 therapy for hypothyroidism •Recovery phase after Non-Thyroidal illness
Raised	Decreased	Decreased	•Chronic Autoimmune Thyroiditis •Post thyroidectomy, Post radioiodine •Hypothyroid phase of transient thyroiditis
Raised or within Range	Raised	Raised or within Range	•Interfering antibodies to thyroid hormones (anti-TPO antibodies) •Intermittent T4 therapy or T4 overdose •Drug interference- Amiodarone, Heparin, Beta blockers, steroids, anti-epileptics
Decreased	Raised or within Range	Raised or within Range	•Isolated Low TSH -especially in the range of 0.1 to 0.4 often seen in elderly & associated with Non-Thyroidal illness •Subclinical Hyperthyroidism •Thyroxine ingestion
Decreased	Decreased	Decreased	•Central Hypothyroidism •Non-Thyroidal illness •Recent treatment for Hyperthyroidism (TSH remains suppressed)
Decreased	Raised	Raised	•Primary Hyperthyroidism (Graves' disease), Multinodular goitre, Toxic nodule •Transient thyroiditis: Postpartum, Silent (lymphocytic), Postviral (granulomatous, subacute, DeQuervain's), Gestational thyrotoxicosis with hyperemesis gravidarum
Decreased or within Range	Raised	Within Range	•T3 toxicosis •Non-Thyroidal illness

References: 1. Interpretation of thyroid function tests. Dayan et al. THE LANCET • Vol 357 • February 24, 2001
 2. Laboratory Evaluation of Thyroid Function, Indian Thyroid Guidelines, JAPI, January 2011, vol. 59

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Investigation

Vitamin D Total - 25 Hydroxy (OH)
 (Serum, Electrochemiluminescence immunoassay (ECLIA))

Observed Value

27.6

Unit

ng/mL

Biological Reference Interval

Deficiency: < 20
 Insufficiency: 20-30
 Sufficiency: 30-100
 Hypervitaminosis: > 100

Interpretation:

- Vitamin D is a fat soluble vitamin and exists in two main forms as D3 & D2. Both are converted to 25(OH) vitamin D in liver.
- For diagnosis of vitamin D deficiency, it is recommended to have clinical correlation with serum 25(OH) vitamin D, serum calcium, serum iPTH & serum alkaline phosphatase
- During monitoring of oral vitamin D therapy- suggested testing of serum 25(OH) vitamin D is after 12 weeks or 3 months of treatment.

Caution:

- Patients on Biotin supplement may have interference in some immunoassays. With individuals taking high dose Biotin (more than 5 mg per day) supplements, at least 8-hour wait time before blood draw is recommended.

Disclaimer:

- The required dosage of vitamin D supplements & time to achieve sufficient vitamin D levels show significant seasonal (especially winter) & individual variability depending on age, body fat, sun exposure, physical activity, genetic factors (especially variable vitamin D receptor responses), associated liver or renal diseases, malabsorption syndromes and calcium or magnesium deficiency.
- Vitamin D toxicity is known but very rare. Kindly correlate clinically, repeat with fresh sample if indicated.

Associated Tests:

- iPTH-Intact Molecule Parathyroid hormone Serum/Plasma (P0114), Calcium (C0017), Vitamin D plus profile (V0016)

Reference:

- Package insert
- Arch Pathol Lab Med—Vol 141, November 2017

IgE Total antibody

(Serum, Electrochemiluminescence immunoassay (ECLIA))

508

IU/mL

0-100

Interpretation:

- High IgE levels are found in asthma, allergic bronchopulmonary aspergillosis, primary immunodeficiency, parasitic infections, inflammatory diseases and some cancers.
- Low IgE levels may be seen in auto-immune disorders.

Clinical Utility:

- Helps in screening and detecting allergies.
- Measures the overall quantity of immunoglobulin E in the blood, not the amount of a specific type.
- Antigen-specific IgE is the next step to identify the responsible allergen.

Note:

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Investigation	Observed Value	Unit	Biological Reference Interval
- Patients on Biotin supplement may have interference in some immunoassays. With individuals taking high dose Biotin (more than 5 mg per day) supplements, at least 8-hour wait time before blood draw is recommended. - This test does not necessarily indicate presence of allergy, to investigate more about allergy – advised Allergen specific IgE tests /panels by method according to age, symptom (Asthma, rhinitis, eczema), environmental and geographical locations. Contact Laboratory for more information on the same.			

Associated Tests:

- Allergy - Adult (Comprehensive) Panel (A0372), Allergy Screening (A0626)

Reference:

- Kit insert
- Heyworth-Smith D, Campbell P. Laboratory Diagnosis of Allergy. QML pathology newsletter, Brisbane, 2017.



FSH (Follicle Stimulating Hormone)
 (Serum, Electrochemiluminescence immunoassay (ECLIA))

40.7

mIU/mL

Normal Menstruating Women
 Follicular Phase : 1.4-9.9
 Mid Cycle Phase : 0.2-17.2
 Luteal Phase : 1.1-9.2
 Post Menopausal : 19.3-100.6
 Note : Change in Method
 Non-Smoking: < 3
 Smoking: < 5

CEA (Carcino Embryonic Antigen), Serum 1.29
 (Serum, Electrochemiluminescence immunoassay (ECLIA))

ng/mL

Interpretation :

- CEA (Carcinoembryonic Antigen), is a tumor associated antigen
- Elevated CEA is noted in adenocarcinoma, especially colorectal cancer, carcinoma of lung, breast, liver, pancreas, prostate, stomach, ovary, lymphatic system and skin.
- Benign conditions which can elevate CEA include smoking, hepatic diseases, infections, inflammatory bowel disease, trauma, autoimmune diseases, renal disorders, pancreatitis, cirrhosis of the liver, peptic ulcer, hypothyroidism, chemotherapy and radiation.

Clinical Utility:

- CEA is used in the monitoring & follow up of patients with colorectal, gastric, breast, lung, prostatic, pancreatic and ovarian carcinoma.
- This test should not be used for diagnosis of cancer in isolation, as both false positive and negatives can occur

Disclaimer

- The above results obtained cannot be compared to or interchanged with results determined by different assays due to differences in assay methods and reagent specificity
- A single test result is difficult to evaluate, but a number of tests, done weeks apart, shows trends in disease progression or regression.

Reference:

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Investigation**Observed Value****Unit****Biological Reference Interval**

- Kit insert
- Greg.L.Perkin. et.al. Serum Tumor Markers. American family physicians sep.2003 vol.68 no.6
- Nicholson BD, Shinkins B, Pathiraja I, Roberts NW, James TJ, Mallett S, Perera R, Primrose JN, Mant D. Blood CEA levels for detecting recurrent colorectal cancer. Cochrane Database Syst Rev. 2015 Dec 10;2015(12):CD011134. doi: 10.1002/14651858.CD011134.pub2. PMID: 26661580; PMCID: PMC7092609.

**HBsAg Screening**

(Serum, Electrochemiluminescence immunoassay (ECLIA))

Non Reactive,0.308

IU/mL

Non Reactive: < 0.90

Reactive: >= 1.0

Borderline: 0.90-1.0

All Reactive results must be confirmed by Neutralizing confirmatory test or by HBV DNA detection assay.

-- End of Report --



MC-2034

Test Marked with NABL symbol are in the scope of accreditation

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MEDICAL LABORATORY REPORT**Sample Collected At:**

Call2 Connect Pvt Ltd, 4th Dimension Raheja Mindspace Malad West Mumbai 400086

Processing Location :

Processing Location Metropolis Healthcare Ltd. Bhandarkar Road, Pune - 411004